ISLAMIC UNIVERSITY OF TECHNOLOGY

Lab Report 10

Livestock Management System

CSE 4408: System Analysis and Design Lab

Lab 10: Process Specifications and Structured Decisions

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1. Introduction and System Context

1.1 Organizational Context

The Livestock Management System is an integrated farm management system for poultry, fisheries and cattle operations. The proposed system centralizes daily operations, feed and inventory control, health monitoring, sales processing and owner reporting. Its main goal is to improve efficiency, reduce wastage, support prompt alerts and provide real-time monitoring for farm decisions.

The earlier DFD work identified the system boundary, five major business processes and the logical data stores required for implementation. Lab 10 extends that analysis by specifying the internal logic of selected primitive processes. The focus shifts from what data moves between entities, processes and stores to how the system validates, transforms and records the data.

1.2 Purpose and Scope of the Report

This report documents the process specification and structured decision analysis for selected primitive processes in the Livestock Management System. It identifies candidate primitive processes from the Level 0 and lower-level DFDs, selects exactly three processes for formal specification, prepares a process specification form for each selected process and represents the process logic using Structured English, a Decision Table and a Decision Tree.

The report uses the same logical data-store numbering used in the DFD presentation: D1 Farm Structure, D2 Operation Logs, D3 Inventory Records, D4 Health Records and D5 Sales Orders. Each specification is written so that it can be traced back to the relevant DFD process number and data flows.

1.3 Role of Process Specifications

A DFD describes external entities, data flows, data stores and process boundaries. A process specification, or mini specification, describes the detailed processing rules inside a primitive process. It reduces ambiguity, validates that the DFD has the required inputs and outputs, and provides a precise implementation blueprint for programmers.

The selected techniques are used according to the nature of the process logic. Structured English is used for a sequential process with one repeated block, a Decision Table is used for several interacting validation conditions, and a Decision Tree is used where the order of decisions and branch outcomes must be visually traceable.

2. Candidate Primitive Processes and Selection

2.1 Candidate Primitive Processes

The candidate processes were taken from the team presentation and the refined DFD set. Four candidates come from the child diagram of Process 2, Record Daily Operations, and one candidate comes from Process 5, Process Sales and Availability.

DFD No.	Primitive Process	Reason for Consideration
2.1	Capture Daily Feed Log	This process starts the daily feed-entry flow and collects the worker-submitted operational data.
2.2	Validate Batch & Quantity	This process contains multiple validation conditions involving livestock group, feed batch, quantity and stock level.
2.3	Generate Task Schedule	This process generates worker tasks from operation status and updated farm information.
2.4	Update Daily Operation Log	This process applies an ordered sequence and repeats feed-line updates before producing an operation summary.
5	Process Sales & Availability	This process decides whether a buyer order can be confirmed, partially fulfilled or rejected based on buyer and stock conditions.

Table 1: Candidate primitive processes considered for formal process specification.

2.2 Selected Processes

Selected Process	Technique	Selection Rationale
2.4 Update Daily Operation Log	Structured English	The process is mostly sequential and contains a simple loop over feed usage lines. Structured English expresses this logic clearly without an unnecessarily large table.
2.2 Validate Batch & Quantity	Decision Table	The process depends on several interacting yes/no conditions. A table shows rule patterns, merged rules, early stopping and validation outcomes compactly.
5 Process Sales & Availability	Decision Tree	The process follows an ordered branch sequence: receive order, validate buyer information, check stock and then choose confirm, partial offer or rejection.

Table 2: Selected processes and structured decision analysis techniques.

2.3 Processes Not Selected

Process 2.1 Capture Daily Feed Log was not selected because most of its work is data capture and required-field collection, which can be handled through form controls and data dictionary rules. Process 2.3 Generate Task Schedule was not selected because its current rules are simpler than the validation and sales-availability logic. These processes remain important in the DFD, but the selected three processes better demonstrate sequence, condition and branching constructs.

3. Process Specification Forms

3.1 Process Specification Form: Update Daily Operation Log

Field	Specification
Process Number	2.4
Process Name	Update Daily Operation Log
Brief Description	Appends accepted feed usage details to the daily operation log, links each line to worker and farm context, calculates total daily feed consumption and generates an operation summary.
Input Data Flows	Validated Feed Log; Task Completion Status
Output Data Flows	Updated Operation Log; Operation Summary
Data Stores Used	D2 Operation Logs; D3 Inventory Records
Trigger	Farm worker feed log is accepted after validation.
Precondition	Feed batch and livestock group are valid, and the accepted feed log contains one or more feed usage lines.
Postcondition	The daily operation record is saved and a summary is available for reporting.
Business Rules	Every feed usage line must be recorded under the correct date, worker, farm section and livestock group. The daily total must be recalculated after accepted entries are appended.
Technique Used	Structured English, because the logic is ordered and includes a small

	repeated block.
Unresolved Issues	The final audit-retention period for corrected operation-log entries should be confirmed by the farm manager.

Table 3: Process specification form for Process 2.4.

3.2 Process Specification Form: Validate Batch & Quantity

Field	Specification
Process Number	2.2
Process Name	Validate Batch & Quantity
Brief Description	Checks whether the submitted feed usage data refers to a valid livestock group and feed batch, contains a positive quantity and has sufficient available stock.
Input Data Flows	Daily Feed Logs; Required Feed Data
Output Data Flows	Validated Feed Log; Update Stock; Exception Notice
Data Stores Used	D1 Farm Structure; D3 Inventory Records
Trigger	Worker submits feed usage data.
Precondition	The submitted feed log contains the required fields for worker, livestock group, feed batch and quantity.
Postcondition	The feed log is accepted for operation-log update, returned for correction or held with an exception notice.
Business Rules	The system accepts feed usage only when the livestock group is valid, the feed batch exists, the quantity is greater than zero and stock is sufficient.
Technique Used	Decision Table, because multiple validation conditions interact and early

	failure rules can be merged.
Unresolved Issues	The exact low-stock alert threshold and approval authority for held logs should be finalized.

Table 4: Process specification form for Process 2.2.

3.3 Process Specification Form: Process Sales & Availability

Field	Specification
Process Number	5
Process Name	Process Sales & Availability
Brief Description	Evaluates buyer orders against buyer details and inventory data to confirm the order, offer partial availability or reject the order.
Input Data Flows	Product Orders; Inventory Data
Output Data Flows	Availability; Order Confirmation; Order Rejection
Data Stores Used	D5 Sales Orders; D3 Inventory Records
Trigger	Buyer submits a product order.
Precondition	Buyer details and product items are available for checking.
Postcondition	The order is confirmed, partially offered or rejected, and the buyer is notified of the outcome.
Business Rules	Availability must be checked before confirming sales. Full stock allows confirmation and reservation. Partial stock allows a partial offer only when the buyer accepts the reduced quantity. No stock or invalid buyer information leads to rejection or correction.
Technique Used	Decision Tree, because the decision sequence and unequal branch lengths are important.
Unresolved Issues	The partial-order acceptance rule should be added to the sales form before implementation.

Table 5: Process specification form for Process 5.

4. Structured English Specification

4.1 Structured English for Process 2.4

The following specification uses capitalized logic keywords and indentation. Underlined terms are data dictionary terms from the DFD and process specification forms.

```
UPDATE Daily Operation Log
RECEIVE Validated Feed Log
READ Current Operation Log
FOR Feeding Date
   IF Operation Log DOES NOT EXIST THEN CREATE NEW Daily Operation
Record     ENDIF
FOR EACH Feed Usage Line IN Daily Feed Log Record
   ADD Feed Usage Entry TO Operation Log
   LINK Feed Usage Entry WITH Worker
   LINK Feed Usage Entry WITH Farm Section
   LINK Feed Usage Entry WITH Livestock Group ENDFOR
CALCULATE Total Feed Consumed FOR THE DAY
STORE UPDATED Operation Log
GENERATE Operation Summary END UPDATE
Daily Operation Log
```

Figure 1: Structured English specification for Process 2.4 Update Daily Operation Log

4.2 Narrative Explanation

The process begins only after the feed log has been validated. It first finds the operation log for the feeding date. If no log exists, the system creates a new daily operation record. The repeated part of the process then handles each feed usage line independently so that every item is tied to the responsible worker, farm section and livestock group. After all lines are appended, the system recalculates the total feed consumed and stores the updated record.

This logic is suitable for Structured English because the main path is sequential and the only iteration is the repeated update for each feed usage line. A decision table would be larger than necessary, and a decision tree would not show the repeated data-processing block as clearly.

5. Decision Table

5.1 Decision Table for Process 2.2

The decision table below represents the validation logic for Process 2.2 Validate Batch & Quantity. The dash symbol means "do not care"; once a prerequisite fails, later conditions do not affect the outcome of that rule.

Conditions / Actions	**R1**	**R2**	**R3**	**R4**	**R5**
Livestock group valid?	N	Y	Y	Y	Y
Feed batch exists?	-	N	Y	Y	Y
Quantity greater than zero?	-	-	N	Y	Y
Stock sufficient?	-	-	-	N	Y
Reject: invalid livestock group	X				
Reject: unknown feed batch		X			
Return for quantity correction			X		
Hold log and create stock exception				X	
Accept validated feed log					X
Update inventory record					X

Table 6: Decision table for Process 2.2 Validate Batch & Quantity.

5.2 Rule Interpretation

Rule R1 rejects the feed log immediately when the livestock group is invalid. Rule R2 rejects the log when the livestock group is valid but the feed batch is unknown. Rule R3 returns the log for correction when the quantity is not greater than zero. Rule R4 holds the log and creates a stock exception when the requested quantity exceeds available stock. Rule R5 accepts the log and updates the inventory record only when all four validation conditions are satisfied.

5.3 Simplification and Error Check

The original four yes/no conditions could produce sixteen rule columns. The table is simplified to five meaningful rules by using early prerequisite checks. For example, when the livestock group is invalid, it is unnecessary to check the feed batch, quantity or stock level. No contradictions are present because no identical condition pattern leads to two different actions. The merged rules remove redundant combinations while preserving all possible validation outcomes.

6. Decision Tree

6.1 Decision Tree for Process 5

The decision tree below represents Process 5 Process Sales & Availability. It is drawn from left to right so that the reader can trace the exact order of decisions and resulting actions.

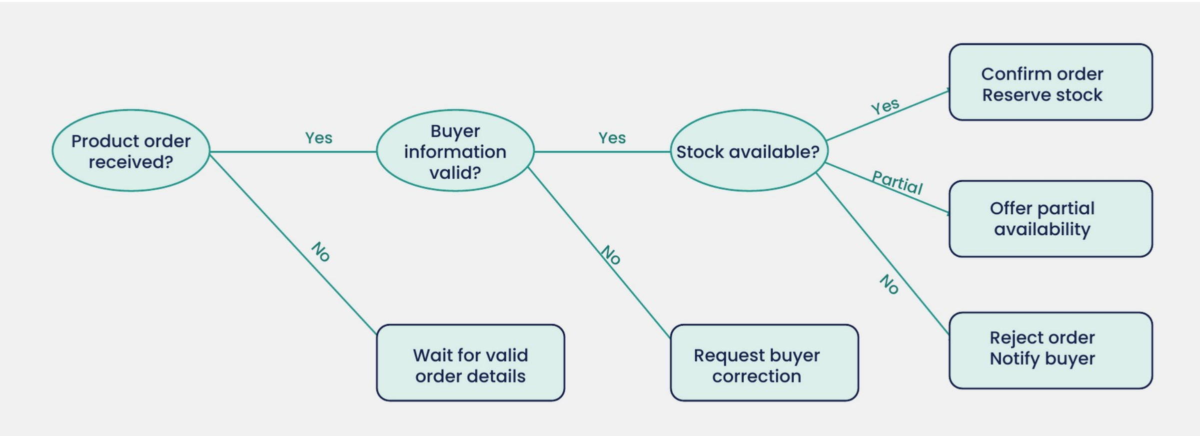

Figure 2: Decision tree for Process 5 Process Sales & Availability.

6.2 Branch and Outcome Explanation

The root condition checks whether a usable product order has been received. If the order is missing or incomplete, the system waits for valid order details. If the order exists, the next condition checks buyer information. Invalid buyer information does not require a stock check because the order cannot proceed until the buyer record is corrected.

Only valid buyer orders reach the stock-availability check. Full stock produces order confirmation and stock reservation. Partial stock produces a partial availability offer. No stock produces rejection and buyer notification. The tree is appropriate because the sequence is important and the branches are asymmetric; some paths finish after one or two checks, while valid orders require a final inventory decision.

7. Ambiguity Handling and Traceability

7.1 Ambiguities and Current Assumptions

Issue Found	Current Assumption	Next Action
Exact low-stock threshold is not finalized.	A stock exception is created when requested feed quantity exceeds available stock.	Confirm threshold values with the farm manager and update inventory rules.
Partial order policy is unclear.	Partial fulfillment is allowed only when the buyer accepts the reduced quantity.	Add a buyer confirmation rule to the sales form and order workflow.
Approval authority for rejected or held feed logs is not finalized.	A farm manager reviews held or rejected transactions.	Define approval authority, audit trail and escalation timing.

Table 7: Ambiguities, assumptions and follow-up actions.

7.2 Traceability to DFDs

Artifact	DFD Process	Inputs	Outputs	Data Stores
Structured English	2.4 Update Daily Operation Log	Validated Feed Log; Task Completion Status	Updated Operation Log; Operation Summary	D2 Operation Logs; D3 Inventory Records
Decision Table	2.2 Validate Batch & Quantity	Daily Feed Logs; Required Feed Data	Validated Feed Log; Update Stock; Exception Notice	D1 Farm Structure; D3 Inventory Records
Decision Tree	5 Process Sales & Availability	Product Orders; Inventory Data	Availability; Order Confirmation; Order Rejection	D5 Sales Orders; D3 Inventory Records

Table 8: Traceability from structured analysis artifacts to DFD processes.

7.3 Formatting and Numbering Check

All tables and figures are numbered sequentially and include captions. The report uses A4 page layout, 12-point body text, 1.5 line spacing and one-inch margins. Each major section is titled and includes a concise narrative so the process specification and decision analysis work can be understood without referring to class notes

8. Conclusion

This report completes the Lab 10 process specification and structured decision analysis for the Livestock Management System. Five candidate primitive processes were identified from the DFDs, and three were selected to cover the required logic types. Process 2.4 Update Daily Operation Log demonstrates sequential processing with iteration through Structured English. Process 2.2 Validate Batch & Quantity demonstrates interacting validation rules through a simplified decision table. Process 5 Process Sales & Availability demonstrates ordered branching through a decision tree.

The completed artifacts clarify the internal business rules behind the DFD processes, identify unresolved assumptions, and provide a code-ready reference for implementation. The analysis also confirms that the selected processes are traceable to the presentation DFDs, input/output flows and logical data stores.